

Overtraining and Training Load Management

Reference notes on distinguishing normal training fatigue from overreaching and overtraining. Paraphrased from sports-science reviews on training load monitoring.

Functional overreaching vs. overtraining

A short period of accumulated fatigue after a deliberately hard training block (functional overreaching) is a normal and often useful part of progress, typically resolving within days to about a week of easier training. Non-functional overreaching and true overtraining syndrome happen when this fatigue is not allowed to resolve and builds up over weeks, leading to a persistent drop in performance and wellbeing.

Common signs of excessive training load

- Unexplained fatigue lasting more than roughly two to three weeks.
- Declining performance despite maintaining or increasing training volume.
- Resting heart rate persistently elevated above the person's normal baseline.
- HRV trending downward over multiple weeks rather than just a couple of days.
- Ongoing sleep disruption.
- Increased irritability, low motivation, or mood disturbance.
- More frequent minor illnesses or infections, reflecting reduced immune resilience.

Why trends matter more than single-day readings

A single hard day followed by one night of suppressed HRV or elevated resting heart rate is expected and not a cause for concern. The pattern only becomes meaningful for an overtraining assessment when it persists across many consecutive days or weeks despite attempts at recovery.

Practical response guidance for the assistant

If a person's data shows a short-term dip (a day or two), the assistant can suggest an easier day or two. If the data shows a persistent multi-week downward trend across several markers at once (sleep, HRV, resting heart rate, and subjective fatigue), the assistant should recommend a longer deload period and suggest checking in with a coach or healthcare provider, since this pattern is outside what routine day-to-day adjustments can fix.